

Tutorial 08

CS105.3 Database Management Systems

Question 01

Consider the following scenario.

Suppose you are tasked with designing a relational database for a travel related system.

1. Draw an Entity Relationship (ER) diagram to represent the above database, stating clearly any assumptions made.
2. Design a relational schema representing these entities and their relationships. Include primary keys, foreign keys, and any necessary attributes.

MyDream Tours Ltd, a company engaged in organizing adventure tours, wishes to develop a database to store information about its tours, customers, partner hotels, tour guides and tour equipment. A customer can book several tours. Each tour is coordinated by a tour coordinator. A tour coordinator organizes several tours at a time. The coordinator is also responsible for making reservations in their partner hotels for their customers to stay during their tours. A coordinator books several hotels at a time and the same hotel can be booked by many coordinators. A guide is assigned to one tour at a time and each tour has only one guide. Customers who have booked tours can also rent tour equipment such as camping gear, boots, safety helmets etc. A customer can rent several pieces of equipment and the same type of equipment can be rented by many customers.

The database will keep track of the following data:

For each Tour:

Tour ID (unique), Tour Name, Category, Tour Date, Tour Location, Rate and Maximum No of Tour Members allowed.

For each Hotel:

Hotel Number (unique), Name, Address, Telephone No (many for each hotel), Number of Rooms, Room Rent and Facilities Available (many for each hotel).

For each Customer:

Reservation ID (unique), Customer Name, Address, Telephone Number, No. of Guests Accompanying and Tour preference.

For each Tour Guide:

Employee ID (unique), Name, Address, Telephone No, Date of Birth, Employee Start Date and Special Training Areas (a guide can have many).

For each Tour Coordinator:

Coordinator ID (unique), Name, Address, Telephone No. and Qualifications (a coordinator can have many).

For each Equipment:

Item Code (unique), Description, Rent Rate and Quantity in Hand.